

Rudrapriya Rajesh Choudhari

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GitHub: github.com/Rudra-CDRI

Portfolio: <https://rudra-cdri.github.io/index.html>

CGPA: 8.42/10



Education

Year	Degree/Examination	University/Board	%/CGPA
2028	DD in Mechanical Engg (Sp. System Design)	Indian Institute of Technology Bhubaneswar	8.42/10
2023	Intermediate (12 th standard)	Board of Intermediate Education, A.P	90%
2021	Secondary School Certificate (10 th standard)	Central Board of Secondary Education	92.17%

Projects

Gesture-Controlled Robotic Arm Teleoperation

ROS 2, PyTorch, MediaPipe, OpenCV

- Developed a real-time teleoperation system for a simulated OpenManipulator-X robotic arm using hand gestures and computer vision in ROS 2 Humble and Gazebo.
- Utilized MediaPipe for extracting 21 3D hand landmarks and trained a lightweight PyTorch neural network (MLP) to map hand poses into a gesture vocabulary for velocity-mode control.
- Implemented a ROS 2 control node utilizing inverse kinematics (ikpy) to convert gestures into joint commands, featuring dual EMA filters for jitter suppression, radial workspace clamping, and joint velocity limiting.

Ionic Wind Generator

Ongoing | Supervisor: Dr. Venugopal Arumuru, IIT Bhubaneswar

- Designing a solid-state electrohydrodynamic (EHD) thruster to generate directed airflow with no moving parts, targeting miniaturized cooling applications.
- Conducted literature survey to identify high-voltage DC power supply constraints; successfully identified and integrated a suitable source to drive corona discharge between emitter and collector electrodes.
- Built and tested a working prototype; measured initial airflow generation and identified design bottlenecks limiting flow rate; currently iterating on electrode geometry and inter-electrode spacing to optimize thrust.
- Investigating boundary layer control effects of ionic wind for potential drag-reduction applications.

Krishi Drone — Autonomous Precision Agriculture UAV

e-Yantra Robotics | IIT Bombay

- Contributed to development of an autonomous quadrotor for pest and disease detection using overhead camera-based computer vision; target areas are localised and biopesticide sprayed only on affected regions.
- System architecture built on ROS 2; ROS 2 Nav2 handles precision waypoint following guided by real-time image feedback from a fixed overhead camera.
- Simulation environment validated in Gazebo before deployment on real hardware, with control system designed to handle both platforms through a unified software interface.

CFD Analysis of CI Engine Piston Head Geometries

ANSYS Forte | IIT Bhubaneswar

- Performed reactive-flow CFD simulations in ANSYS Forte to compare combustion performance of Flat-top and Circular Dish piston geometries in a CI engine (Bore: 100 mm, Stroke: 100 mm).
- Configured RANS RNG $k-\varepsilon$ turbulence model with CHEMKIN-Pro chemistry solver; set isothermal wall BCs and a swirl ratio of 0.3 with injection at -15° BTDC through a 150- μm nozzle.
- Dish-type geometry produced peak turbulence velocity of 750 cm/s vs. 400 cm/s (flat), peak in-cylinder pressure of 1.5×10^8 dyne/cm², and peak temperature of 1550 K, confirming superior squish-driven air-fuel mixing and near-complete combustion.

2DOF Robotic Manipulator — Kinematics & Trajectory Planning

MATLAB Simulink & Simscape

- Designed a CAD model of a 2-DOF planar robotic arm using Simscape Multibody and exported its Rigid Body Tree to implement Forward Kinematics for real-time end-effector pose estimation.
- Implemented Inverse Kinematics using the iterative Levenberg-Marquardt numerical solver to accurately compute joint angles required for the end-effector to trace geometric waypoints (e.g., a square).

- Generated smooth Cartesian paths using Trapezoidal Velocity Profiles to govern acceleration/deceleration, and extracted the manipulator's Jacobian matrix to perform continuous Velocity Kinematics analysis.

Technical Skills

- **Simulation & Design:** ANSYS Forte, SolidWorks, MATLAB, MATLAB Simulink, Simscape, MuJoCo
- **Robotics & Programming:** ROS 2, Gazebo, Isaac Sim, Isaac Lab, Python, C++, PyTorch, Nav2, Git/GitHub, Linux
- **Machine Learning:** Machine Learning, Reinforcement Learning
- **Computer Vision:** OpenCV, MediaPipe

Certifications & Training

ROBOAI — Industrial Training Program

My Equation

Robotics, Artificial Intelligence & Automation

- Completed a 45-day industry-oriented training covering robot mathematics, mechanical CAD, analysis & prototyping, electronics/IoT, ML/AI, ROS 2, Gazebo and OpenCV.
- Applied concepts through hands-on robotics projects.

Web Development Training Program

TeachNook, in association with IIT Roorkee

- HTML, CSS, JavaScript

Relevant Coursework

- **Robotics & Control:** Robot Kinematics & Dynamics (DH Parameters, Transforms, Workspace Analysis, Path Planning), Systems & Controls (State-Space Methods, PID, Laplace/Fourier, Root Locus, Bode/Nyquist, MATLAB-Simulink)
- **Design & Mechanical Systems:** Machine Design & Practice (Tolerances, Fits, Fatigue, Fracture Mechanics, Engineering Drawing, CAD, Introductory FEA with ANSYS), Theory of Machines, Solid Mechanics, Fluid Mechanics, Thermodynamics, Manufacturing Technology
- **Mathematics & Computation:** Linear Algebra, Numerical Methods, Probability & Statistics

Positions of Responsibility

Career Development Cell, IIT Bhubaneswar

Academic Year 2026–27

Student Internship Coordinator (ongoing)

- Coordinating internship outreach and student-industry interface for the institute's Career Development Cell.

Aaroh — Music Society, IIT Bhubaneswar

Academic Year 2025–26

Governor

- Managed annual budget planning, event scheduling, and repertoire selection for the institute's music society.
- Led the band team to **1st prize** in the Inter-IIT 8.0 Band Competition.

Booksolver

June 2024 – October 2024

Subject Expert — Physics

- Produced detailed video solutions for advanced-level Physics problems, emphasizing conceptual clarity and exam strategy.
- Developed structured content on problem-solving approaches for competitive exam preparation.

Extra-Curricular Activities

- Guitarist and Bassist for the institute's band; active participant in cultural events at IIT Bhubaneswar.
- Member of the Chess Club, IIT Bhubaneswar.
- Recreational sports: Table Tennis, Badminton, Swimming.